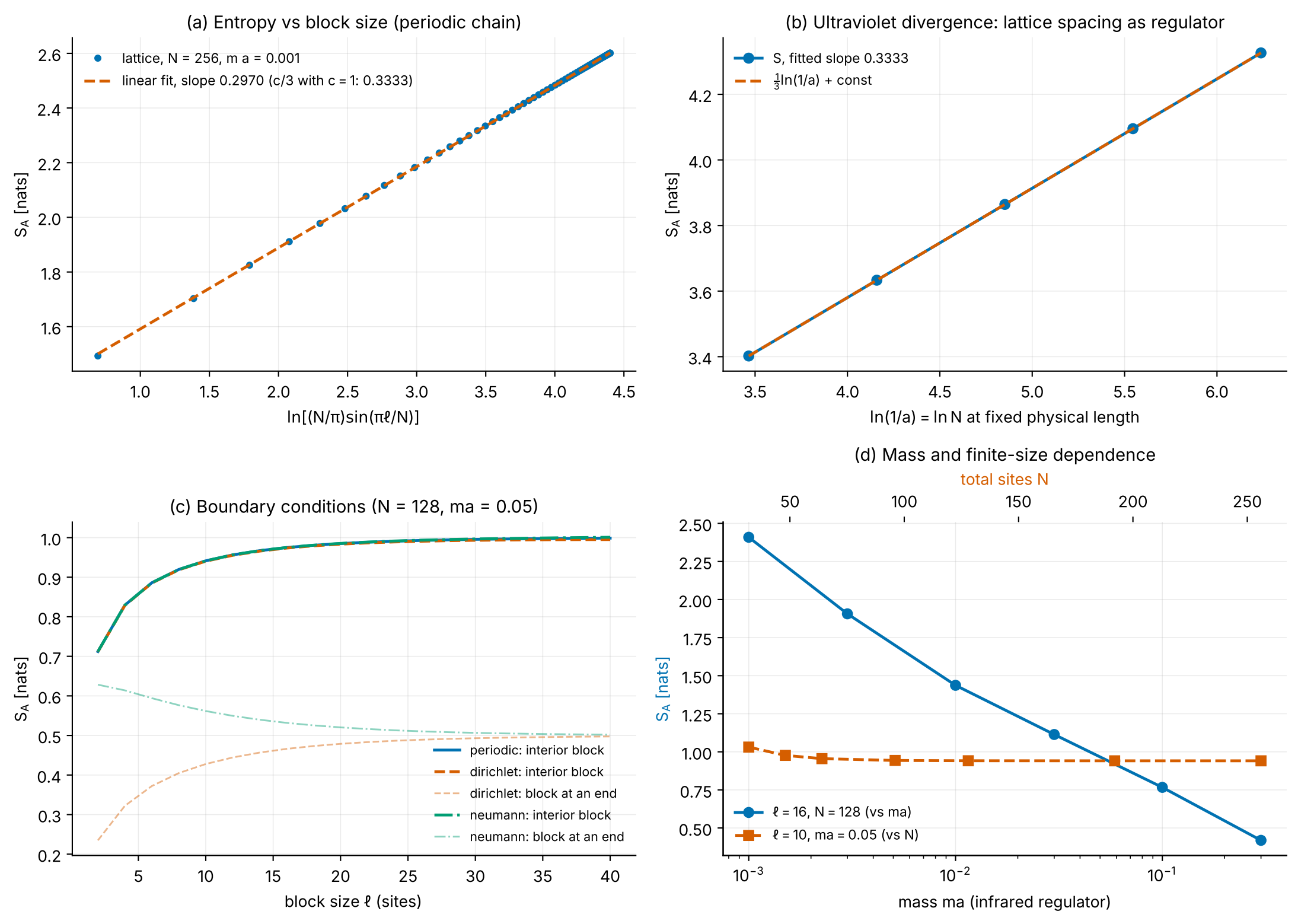
(a) Entropy vs block size (periodic chain)
lattice, N = 256, m a = 0.001
linear fit, slope 0.2970 (c/3 with c = 1: 0.3333)
S_A [nats]
ln[(N/π)sin(πℓ/N)]
(b) Ultraviolet divergence: lattice spacing as regulator
S, fitted slope 0.3333
1/3 ln(1/a) + const
S_A [nats]
ln(1/a) = ln N at fixed physical length
(c) Boundary conditions (N = 128, ma = 0.05)
periodic: interior block
dirichlet: interior block
dirichlet: block at an end
neumann: interior block
neumann: block at an end
S_A [nats]
block size ℓ (sites)
(d) Mass and finite-size dependence
total sites N
ℓ = 16, N = 128 (vs ma)
ℓ = 10, ma = 0.05 (vs N)
S_A [nats]
mass ma (infrared regulator)